

in Indonesia. Alternatively, the **exchange of resources and information** allowed by modern humans' language and symbol use, and their planned and flexible technologies made *Homo sapiens* better able to withstand climatic downturns than Neanderthals.

Others believe that these behaviors were not unique to modern humans. Hominins would have needed to **use rafts or boats** to reach the island of Flores in Indonesia by 800,000 YA. Some late Neanderthal sites also contain elements of technologies normally associated with *Homo sapiens*, although it is possible that Neanderthals may have copied, traded with, or even stolen from modern humans.

A combination of environmental unrest and increased competition is currently considered to be the most likely explanation for **Neanderthal extinction**.



KEY ● Neanderthal sites
● Modern human sites

Neanderthal and human ranges
Modern humans and Neanderthals coexisted for several thousand years. Sites appear to show evidence of interaction between the groups.



THE MAXIMUM EXTENT OF THE LAST ICE AGE

European climates after 23,000 BCE grew steadily cooler, and during the "Last Glacial Maximum" (21,000–18,000 YA), ice caps covered most of northern Europe. Farther south, huge areas of grassland with few trees offered good hunting for groups of humans able to survive the cold.

THE "GRAVETTIAN" CULTURE OF

Europe and Russia (28,000–21,000 YA) is known for its elaborate sites, which often have complex structures and burials, as well as large amounts of shell jewelry, and sculpted bone and antler. Also found at Gravettian sites are some of the earliest known clay objects, including some of the famous "**Venus**" figurines. These may have been fertility or religious charms, or part of a system of exchange between social networks across the region as the **Ice Age intensified**.

"Venus" statuette
This figurine from Willendorf, Austria depicts a stylized pregnant or obese female figure.

exaggerated belly



AT THE HEIGHT OF THE GLACIAL

Maximum, when the ice caps were at their maximum extent, people living in more northerly and mountainous areas retreated to "**refuge**" areas such as—in Europe—northern Spain and southwest France, where this period is known as the "**Solutrean**." Globally, **many groups probably died out**, but some held on in more sheltered regions. To survive the harsh conditions, much time and effort was invested in hunting. Weapons include beautifully worked points known as "leaf-points." Although little evidence survives beyond finely worked bone needles, people probably developed **sophisticated clothing** to keep them warm. Perhaps more importantly, hunters would have worked hard to predict and intercept the movements of herds of large animals, ensuring the **hunting success** that was the difference between life and death.

IN EUROPE, SOPHISTICATED BONE and antler points, needles, and harpoons characterize the "**Magdalenian**" technologies that were used to hunt a wide range of species, especially reindeer.

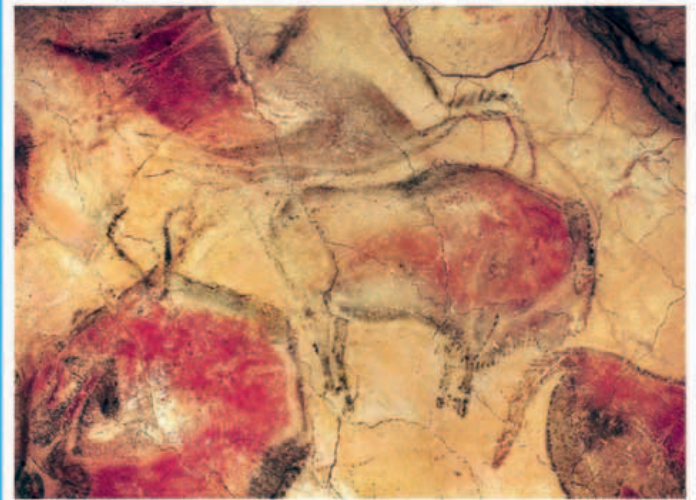
The Magdalenian (18,000–12,000 YA) is famous for its beautiful art objects, engravings, and cave paintings. There are many theories about what these mean and why they were produced. As most depict animals that were hunted, the paintings may represent a magical means of ensuring hunting success, or show information about the best ways to hunt different species. Paintings of imaginary half-human, half-animal creatures and the inaccessibility of some cave art suggest that painting may have been a magical or ritual activity, perhaps practiced by shamans or during initiation or religious ceremonies. Alternatively, paintings

and art objects may have helped establish group identities and territories, as the number of archaeological sites in this period suggests that populations were growing, and competition for rich and localized resources may have been intensifying.

A rise in temperature led to the **retreat of the ice sheets** that had covered northern Europe, and these areas were rapidly recolonized, with **groups expanding** as far north as Siberia by around 14,000–13,000 YA. Some groups later moved on into Alaska and the Americas. Farther east, in China and in the Jomon culture of Japan, some of the first pots manufactured from clay appeared between 18,000 and 15,000 YA.

Altamira cave paintings

This Paleolithic cave painting of bison was discovered at the Altamira cave site in Spain.



32,000 YA Chauvet cave paintings, France

28,000 YA Youngest known Neanderthal sites

28–21,000 YA Gravettian culture

27,000 YA Complex hunter-gatherer sites on the Russian plains

21,000 YA Solutrean technologies appear

21–18,000 YA Last glacial maximum

18,000 YA Magdalenian technologies appear

18,000 YA Date of the controversial Homo floresiensis specimen, "the hobbit"

17,000 YA Earliest known atlatl, or spear-thrower, from Combe Saunière, France

16–15,000 YA Beginning of recolonization of areas of northern Europe abandoned during worst climatic conditions

17,000 YA Lascaux cave paintings, France

15,000 YA Monte Verde, Chile, a South American site with controversially early dates

14,000 YA Earliest domesticated dog